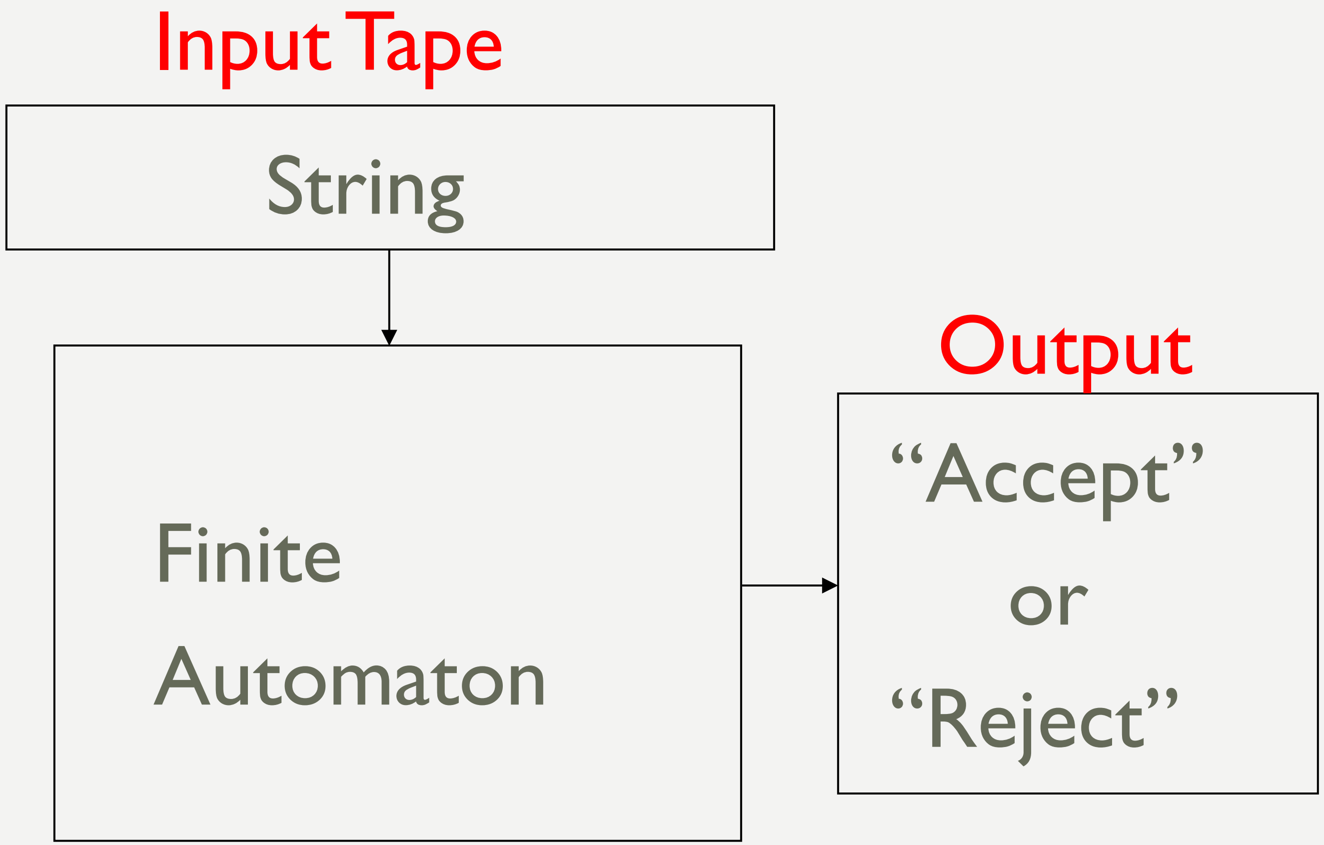
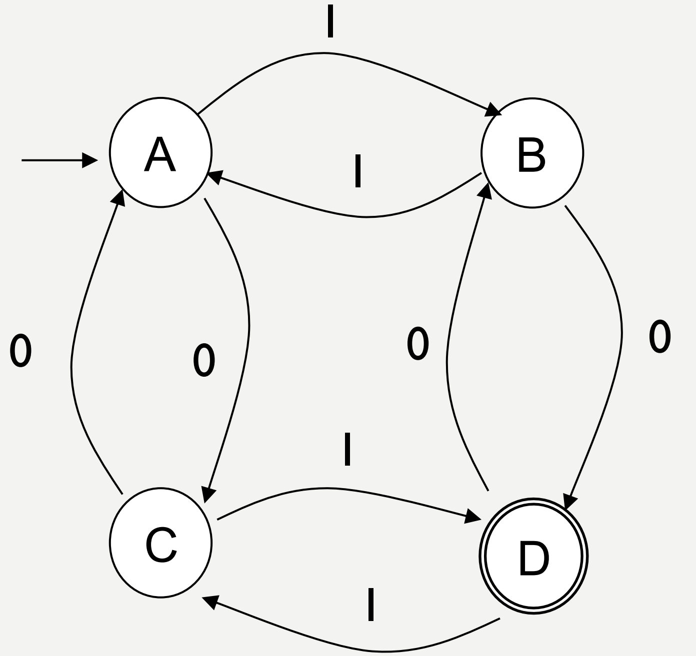
DETERMINISTIC FINITE AUTOMATA

DETERMINISTIC FINITE AUTOMATON (DFA)



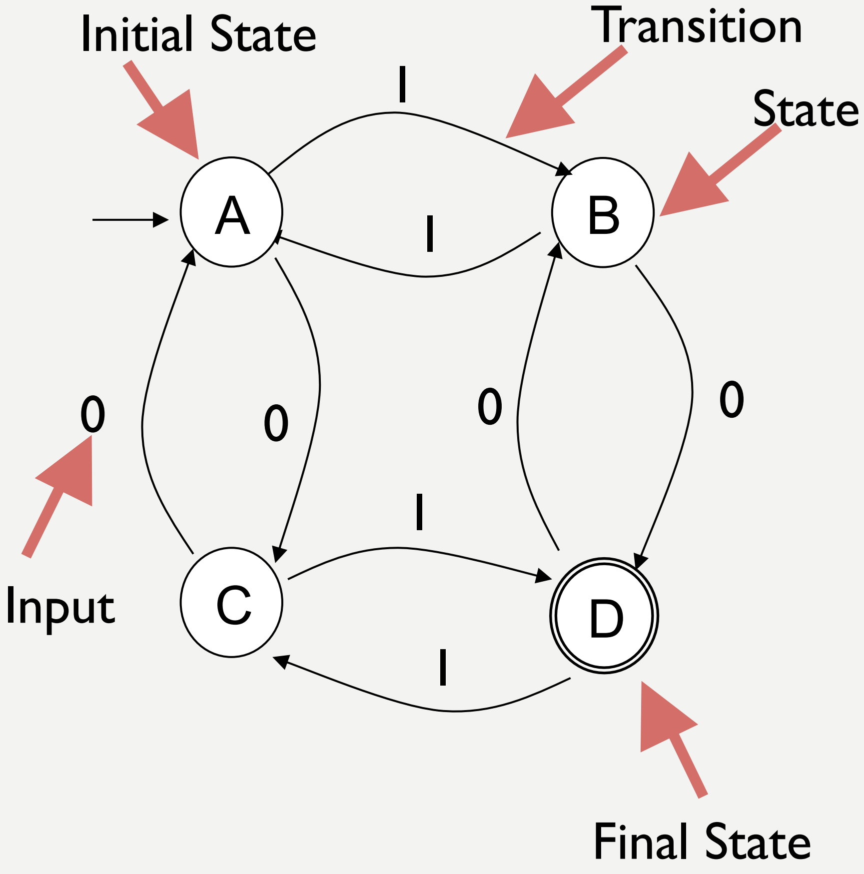
DETERMINISTIC FINITE AUTOMATA

- Determinism
  - Saat berada disuatu state maka state selanjutnya sudah dapat diketahui.
    - Misal saat berada di state A, sudah dapat diketahui jika mendapat input 1 akan lanjut ke state B dan jika mendapat input 0 akan lanjut ke state C.
  - Hanya mempunyai 1 state selanjutnya.
    - Misal saat berada di state A, jika mendapat input 1 hanya akan lanjut ke state B
  - Setiap state mempunyai transisi untuk setiap input dalam himpunan input

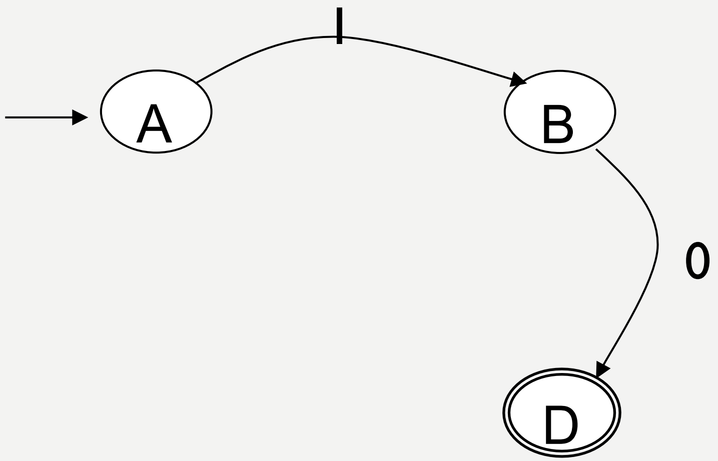


STRUKTUR DFA

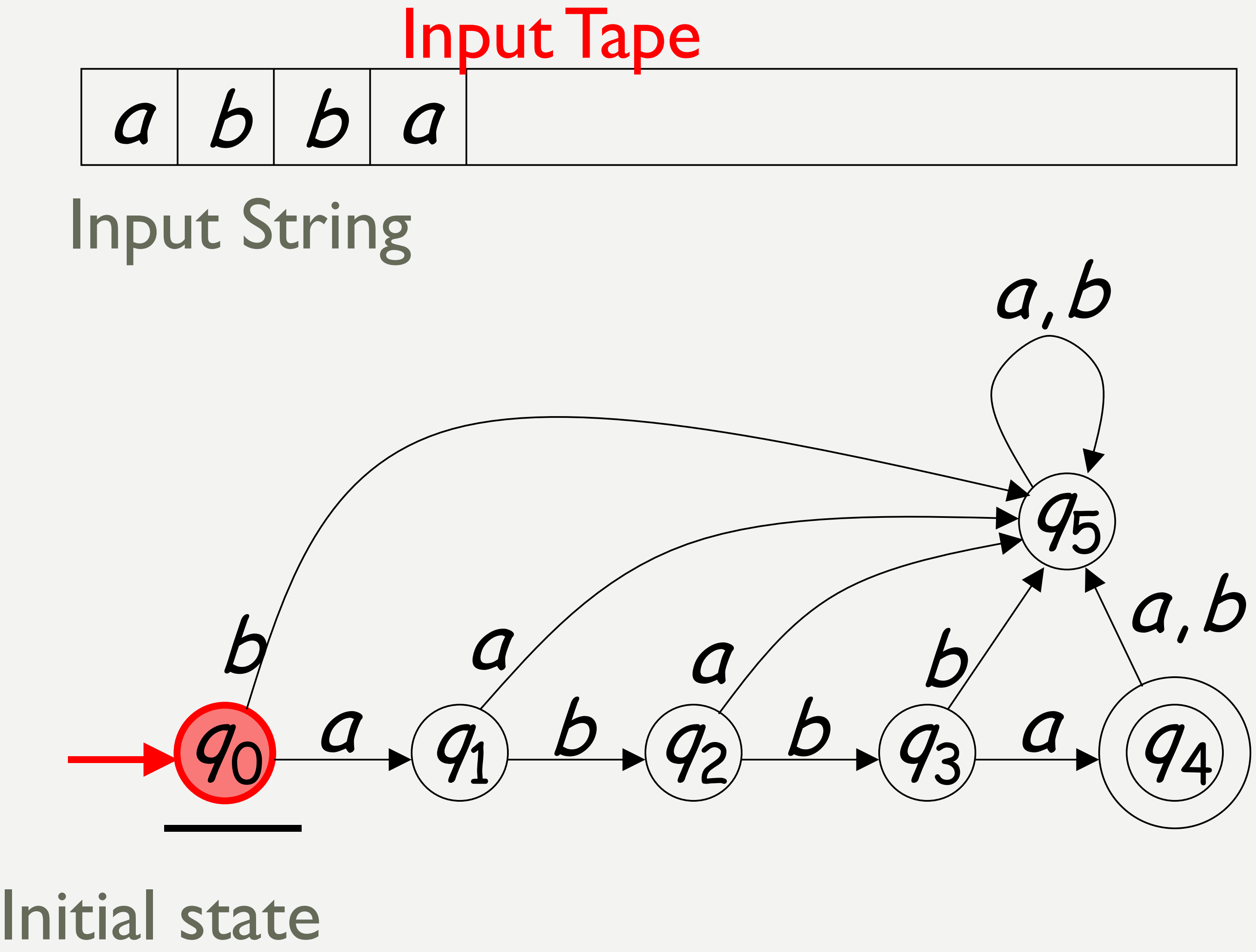
Contoh Struktur DFA



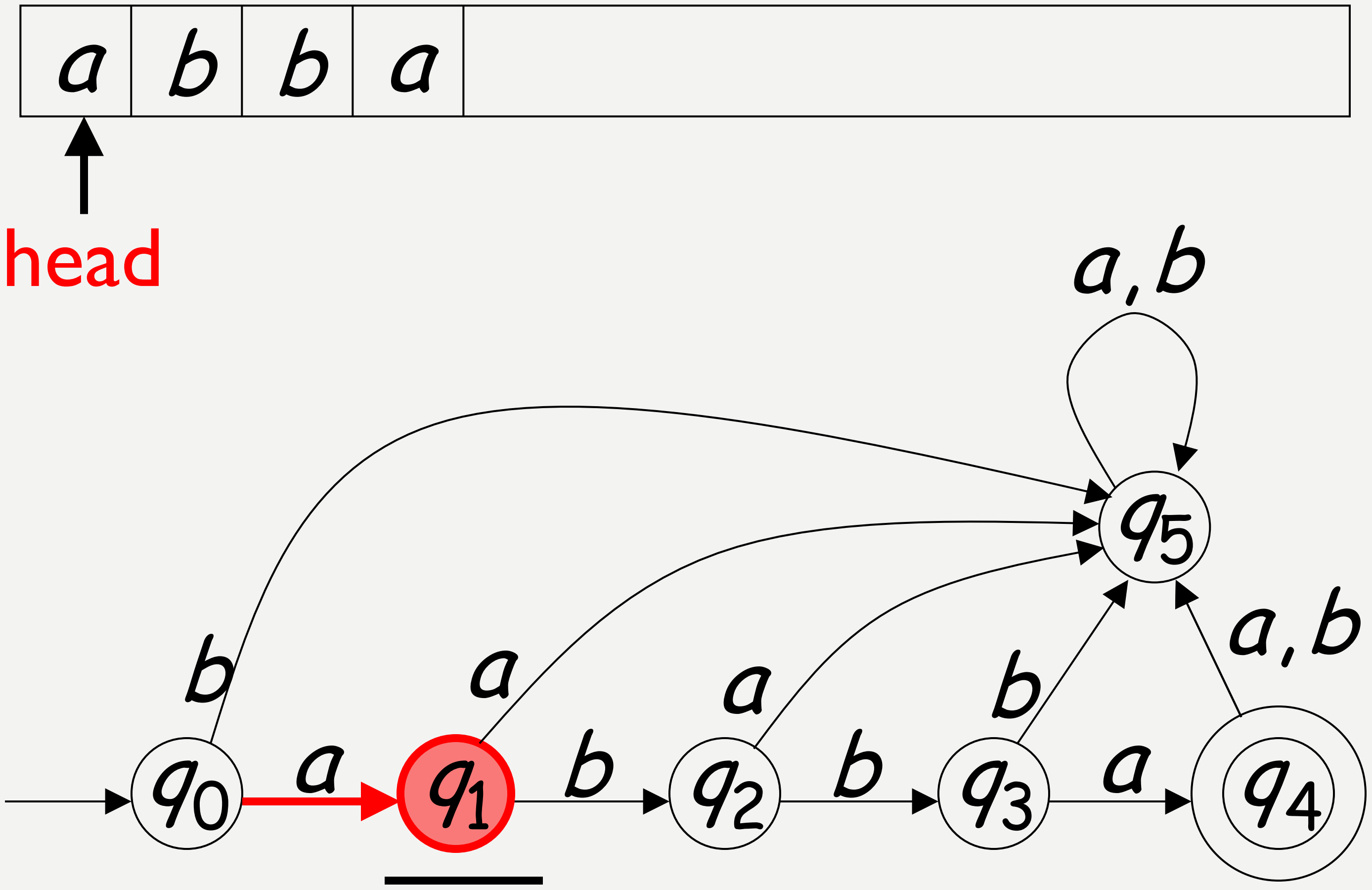
- State awal berada di A
- mendapat input 1
- berpindah ke state B,
- mendapat input 0
- berpindah ke state D yang merupakan final state

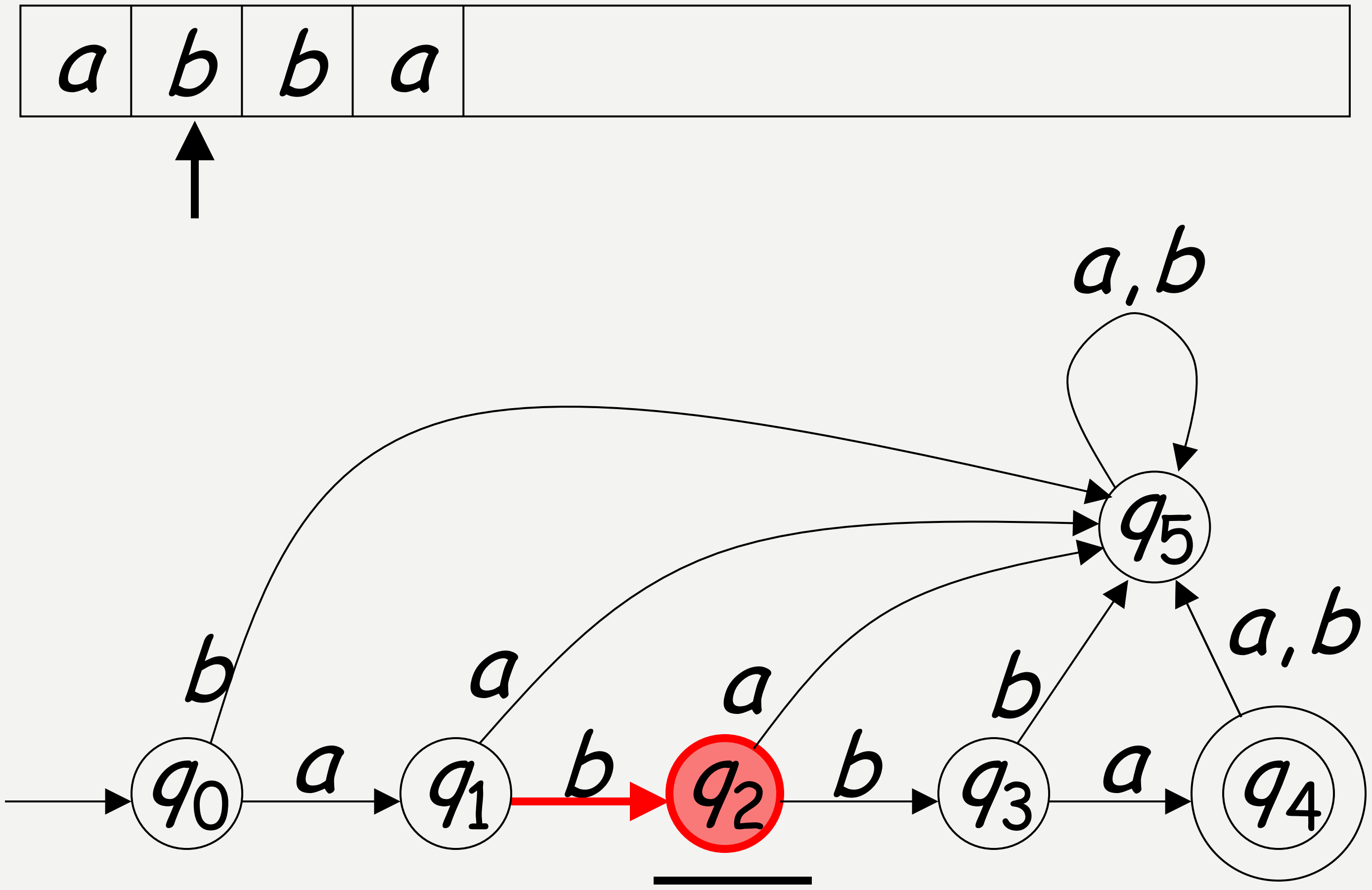


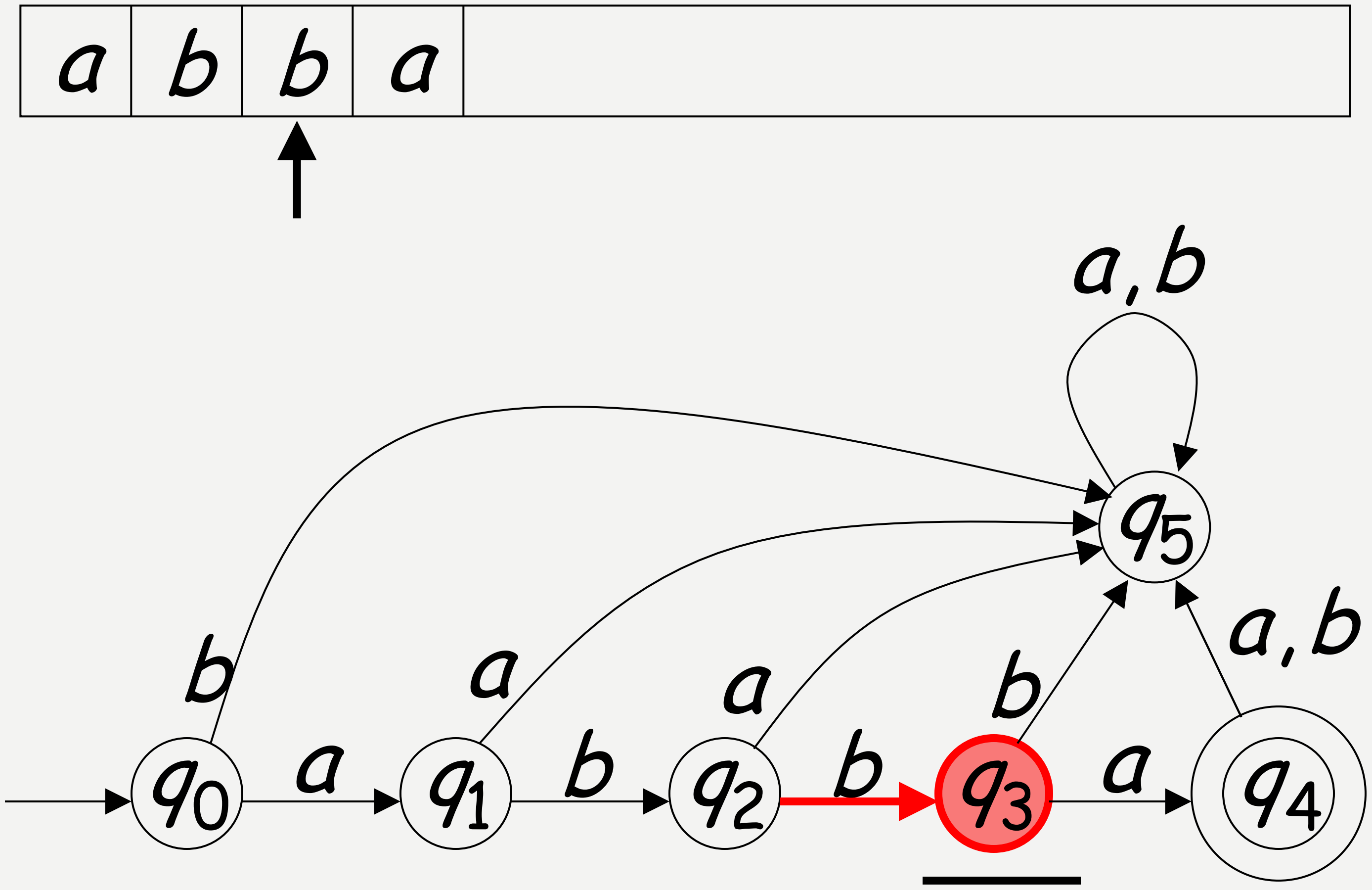
CONTOH


Input Tape
a b b a
Input String
a,b
b
a
a
b
a,b
q_0
a
q_1
b
q_2
b
q_3
a
q_4
q_5
Initial state

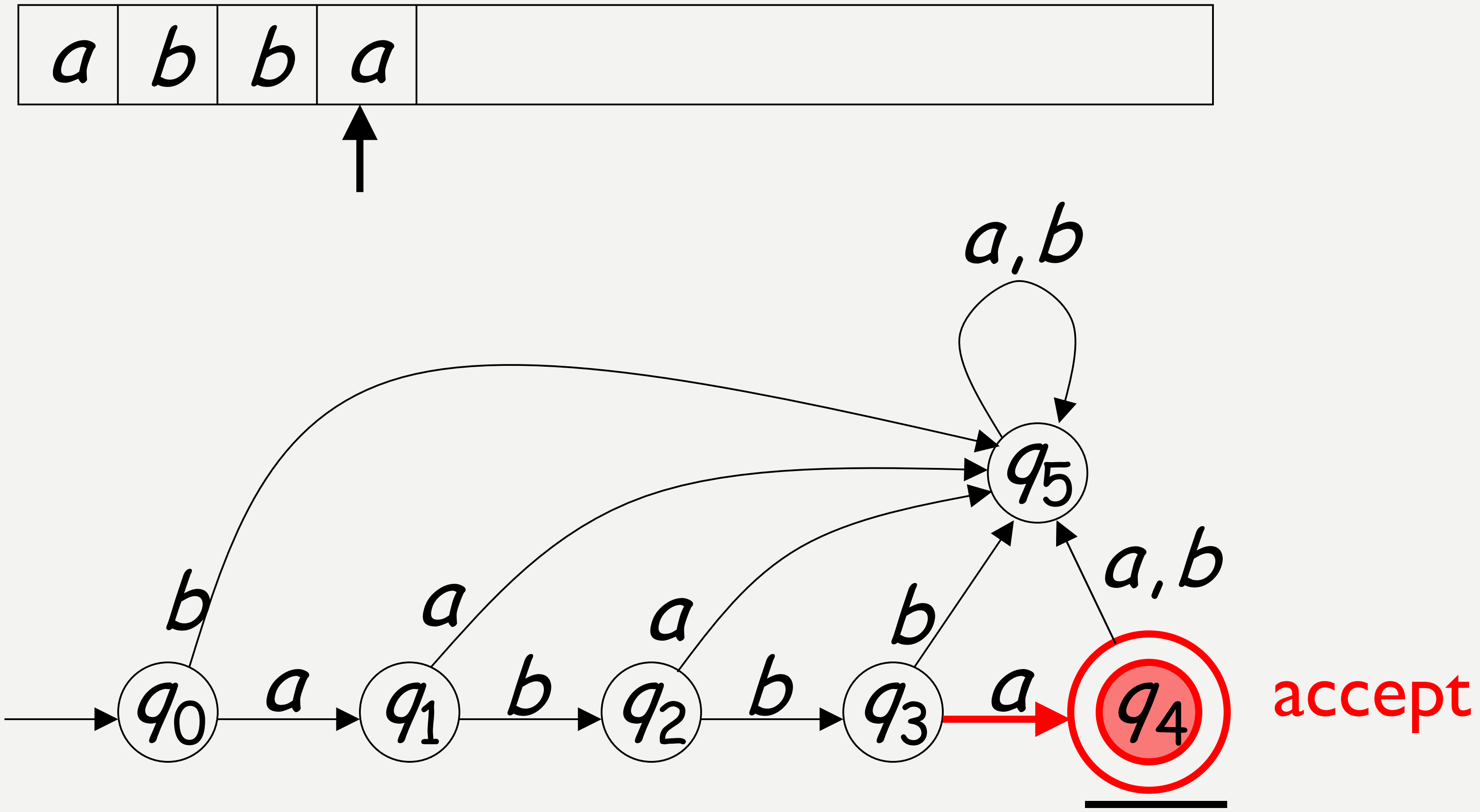
SCANNING THE INPUT




a b b a
a,b
b
a
a
b
a,b
a
b
b
a
q_0
q_1
q_2
q_3
q_4
q_5


a b b a
a,b
b
a
a
b
a,b
q_0
a
q_1
b
q_2
b
q_3
a
q_4
q_5

Input finished

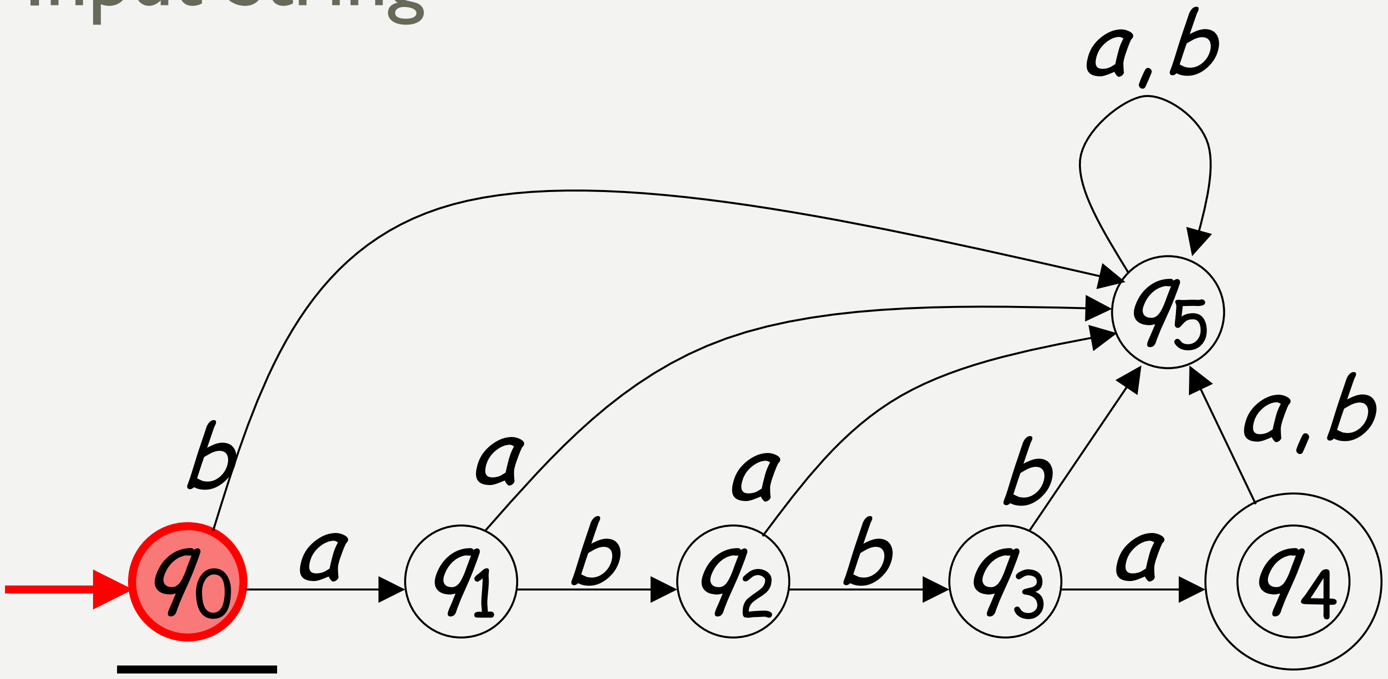


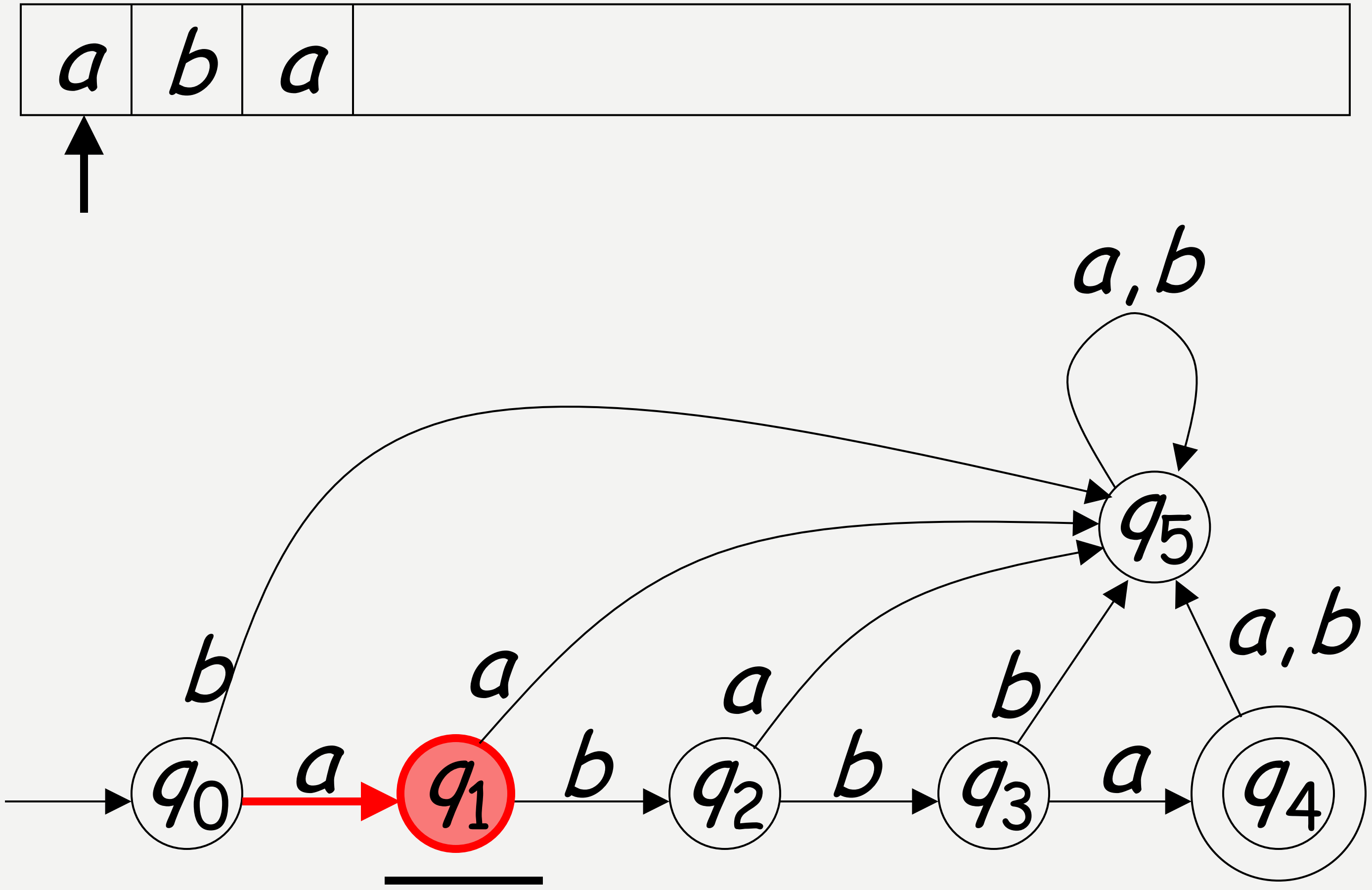
state berakhir pada final state

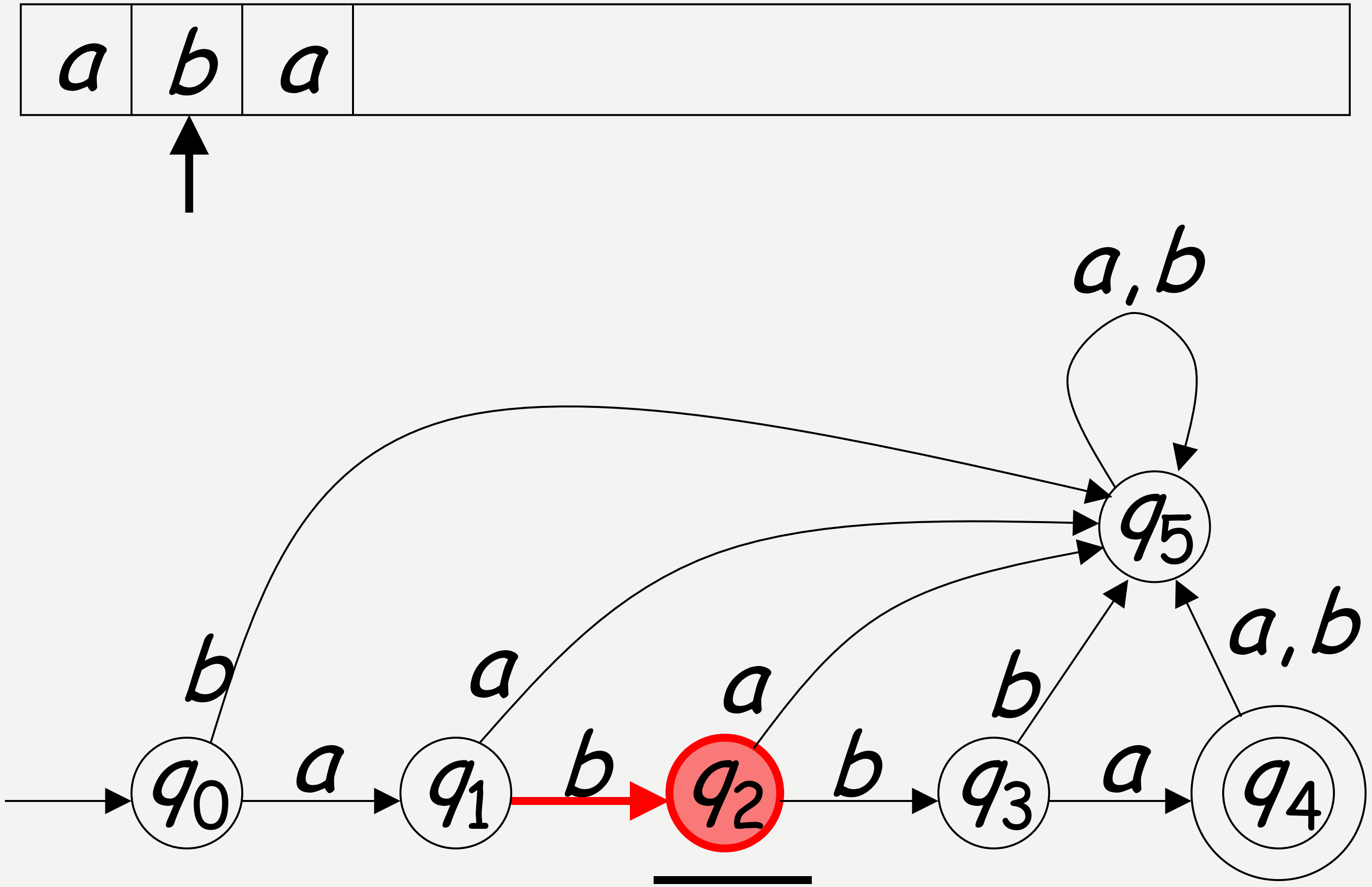
Contoh kasus Reject (1)

a	b	a	

Input String

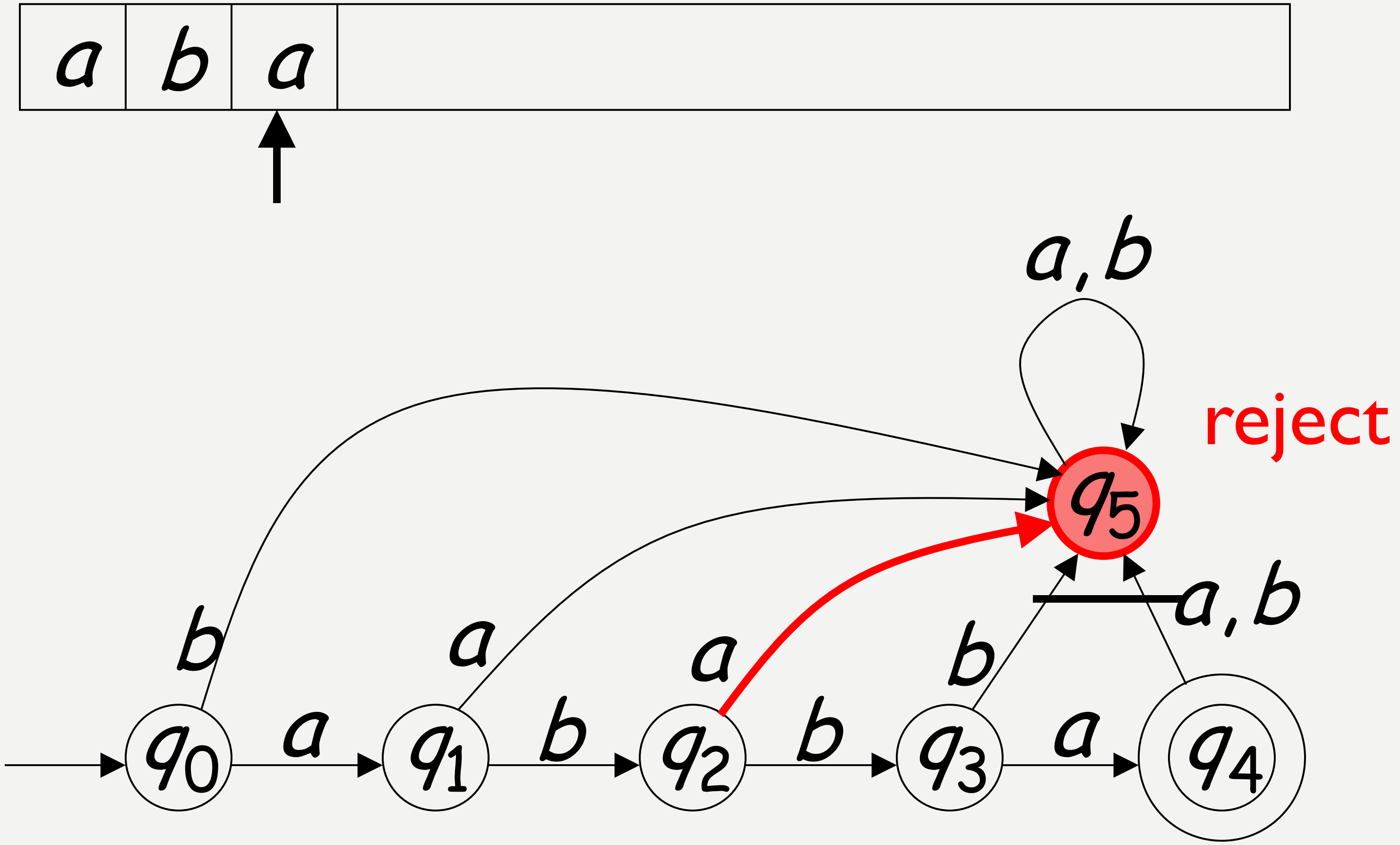



a b a
a,b
b a a b b a
a,b
q_0 q_1 q_2 q_3 q_4 q_5



a
b
a
a,b
b
a
a
b
a,b
a
b
b
a
q_0
q_1
q_2
q_3
q_4
q_5

Input finished



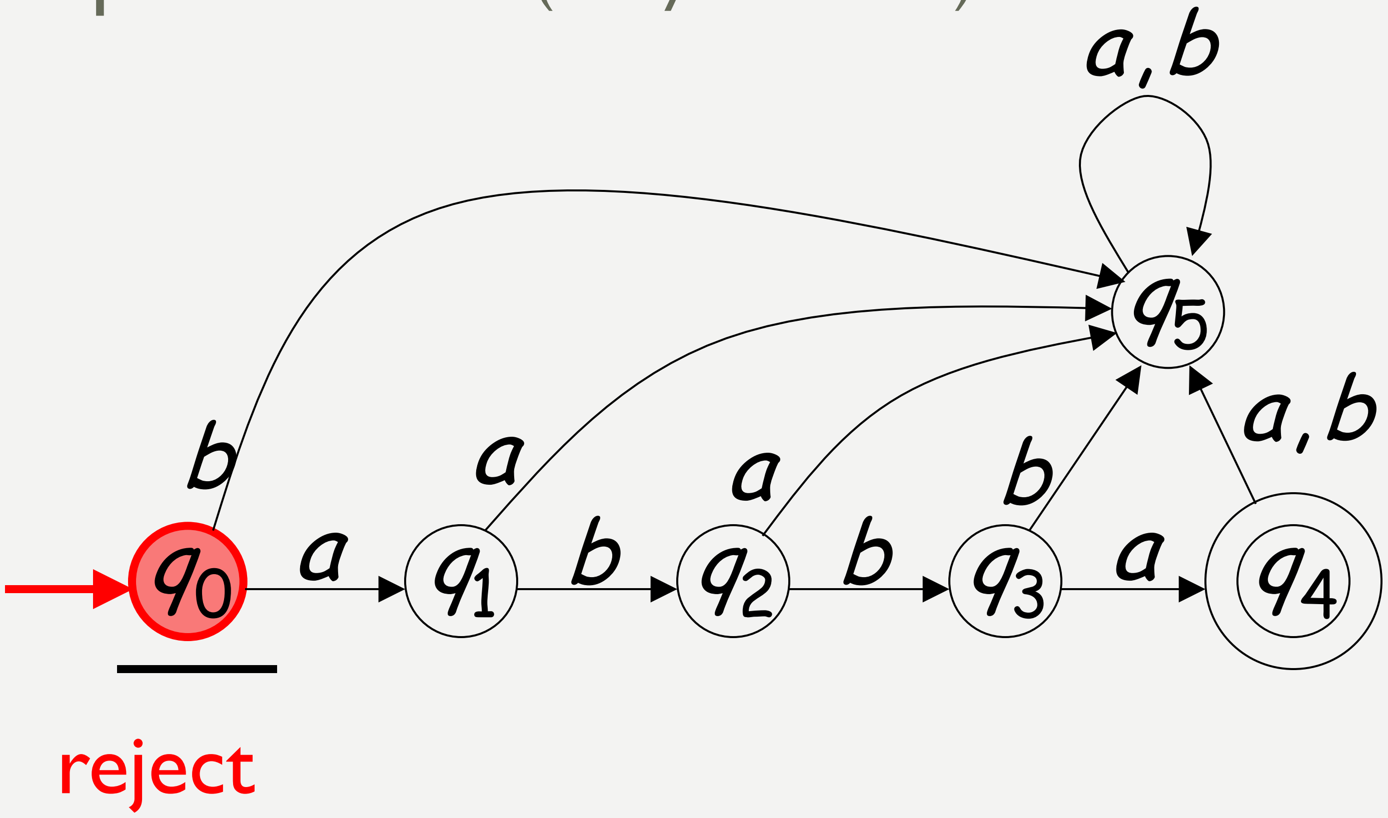
State tidak berakhir pada final state

Contoh kasus Reject (1)

Tape is empty

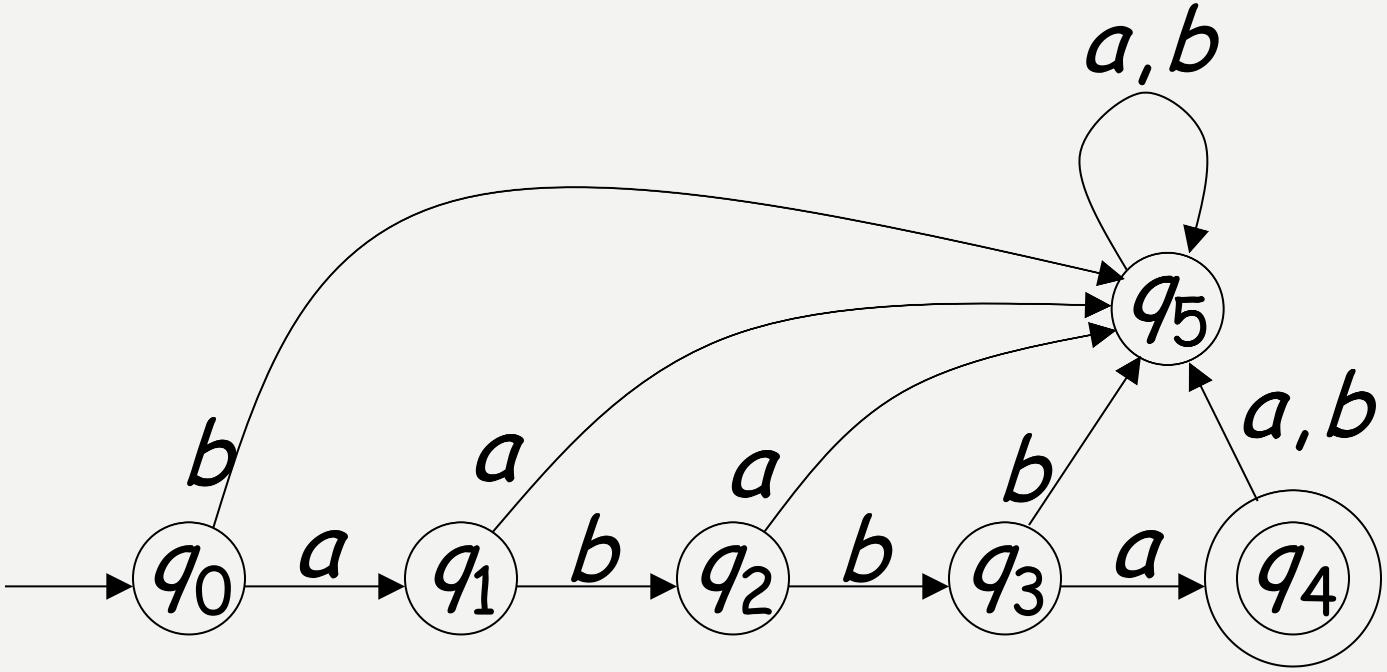
ε

Input Finished (no symbol read)



Automaton hanya menerima satu string

Language Accepted: $L = \{abba\}$



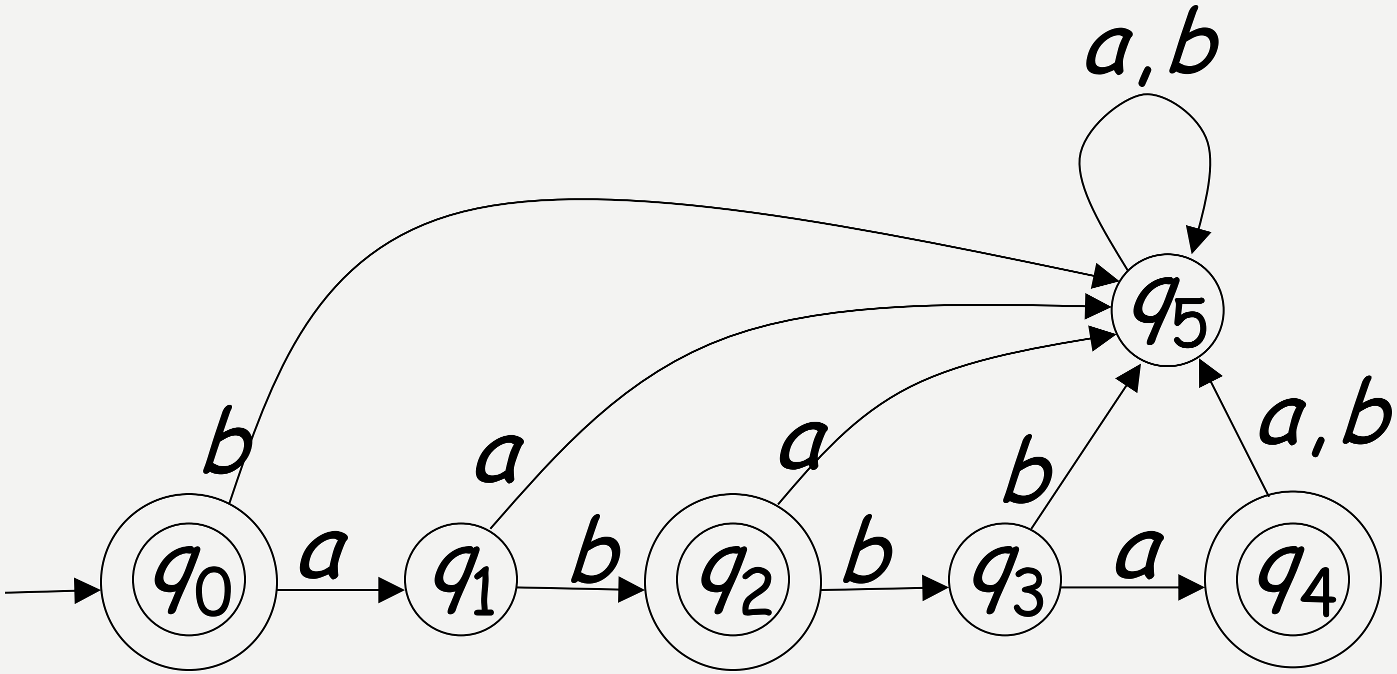
String diterima jika :

Semua input string telah diperiksa
dan state berakhir pada final state

String ditolak jika

Semua input string telah diperiksa
dan state tidak berakhir pada final state

CONTOH DFA DENGAN MULTI FINAL STATE



Accept state

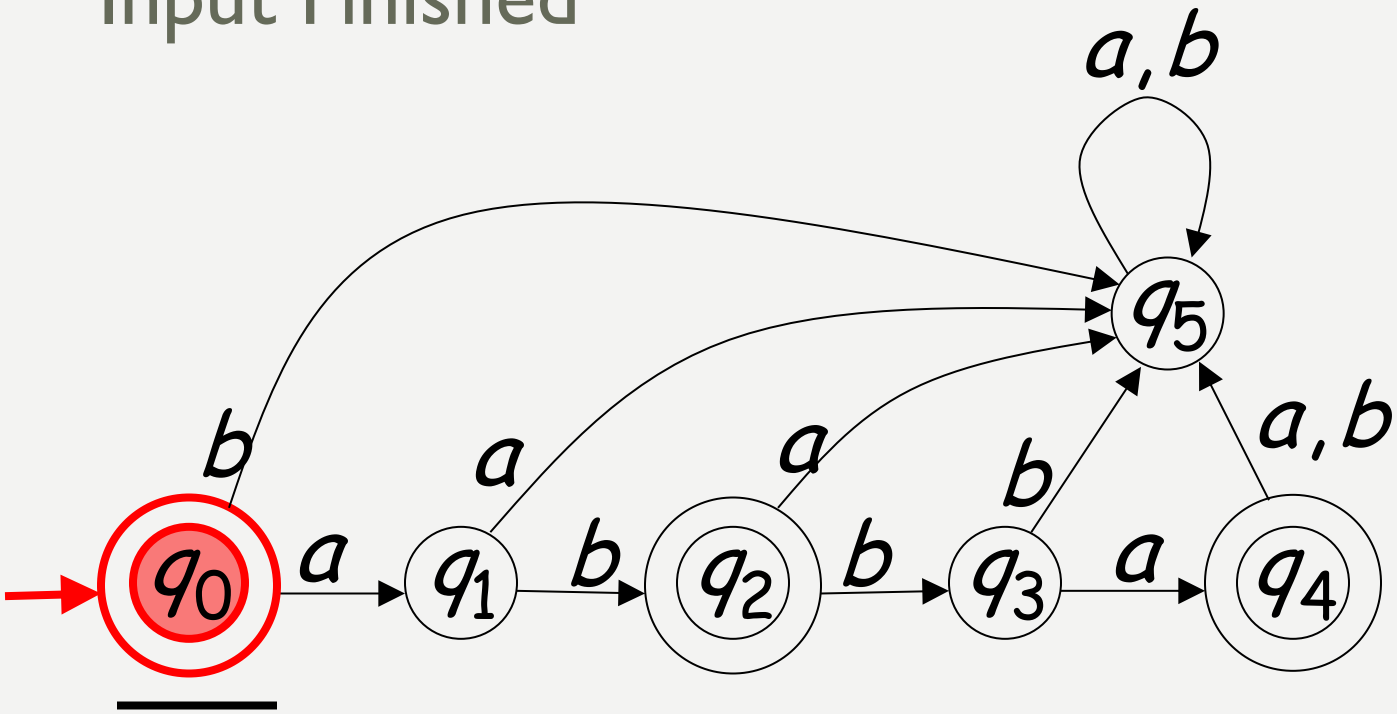
Accept state

Accept state

Empty Tape

ε

Input Finished



$$L = \{\varepsilon, ab, abba\}$$